ABU-SSO User Manual & Installation Guide

Unified Single Sign-On & Digital Campus Wallet

Version: 1.1
Target users: Students, Staff, and System Administrators of Ahmadu Bello University
Course: COEN838 — Advanced Software Engineering

Part A — User Manual (For Students & Staff)

A.1 What is ABU-SSO?

ABU-SSO is a single sign-on system that lets you use **one matric / staff number and password** to access multiple campus services (Student Portal, LMS, Library, Hostel, Wi-Fi, Result Checker) and manage a personal **digital wallet** — all from one dashboard.

A.2 Before You Start

- Use a modern browser: **Chrome, Edge, Firefox, or Safari**.
- You need your **matric number** (students) or **staff number** (staff).
- Your account is created by **ABU ICT** after admission or employment — you do **not** self-register.
- If you're using a shared computer, remember to **log out** when finished.

A.3 Getting Your Account

Unlike consumer websites, **ABU-SSO does not have public self-registration**.

Your situation	What to do
New student / staff member	Wait for ABU ICT to create your account. They will provide your initial password.
Administrator onboarding a user	Log in as admin → Admin Panel → "Create User Account" (see Part C).
You forgot your password	Contact ABU ICT.

Why no self-registration? Because accounts must be tied to real university records. Only the very first user of a fresh deployment becomes an admin automatically; after that, all new accounts come from an administrator.

A.4 Logging In

1. Open your browser and go to the login page:
   `https://abu-sso.onrender.com/login` (production)
   or `http://127.0.0.1:8000/login` (local development)
2. Enter your **Matric / Staff Number** — e.g., `P25EGCP9003` or `ADMIN/001`
3. Enter your **Password**
4. Click **Sign in**

5. On success, you are redirected to your **Dashboard**.
6. On failure, a red error message appears — check your number and password.

> **Tip:** Your matric / staff number is **not case-sensitive**. `p25egcp9003` and `P25EGCP9003` both work.

A.5 Your Dashboard

The dashboard shows everything at a glance:

Section	What it shows
Welcome bar	Your name, role, and matric number
Campus Wallet	Current balance and recent transactions
Digital ID Card	Your name, matric, role, email, and QR code
Campus Services	Six clickable cards — Portal, LMS, Result Checker, Library, Hostel, Wi-Fi
Log out	Ends your session
Admin Panel (admins only)	Blue button to open the admin page

A.6 Using Your Campus Wallet

To check your balance:

- The dashboard shows your current balance at the top of the "Campus Wallet" card.

To top up your wallet:

- Students can request a top-up from library or hostel counters.
- Admins can top up any wallet via the Admin Panel (see Part C).

To pay for a service (e.g., library fine):

- Payments are initiated through the relevant campus service (Library, Hostel, etc.).

To view transactions:

- The dashboard lists the 50 most recent transactions automatically.

A.7 Using Your Digital ID Card

- On the dashboard, your **QR code** appears under "Digital ID Card".
- Present it at library counters and hostel gates for scanning.
- Each QR code encodes **only** `id`, `matric_no`, and `role` — no personal data such as passwords.
- Regenerating your ID card: just refresh the dashboard.

A.8 Using the Six Campus Services (Single Sign-On)

The **Campus Services** card on your dashboard shows six buttons. Clicking any of them opens a **separate campus application** in a new tab — **already signed in** with your ABU-SSO account.

Service	What you can do
Student Portal	Course registration, enrollment
LMS (Moodle)	Lectures, assignments, quizzes
Result Checker	Semester results, GPA
Library	Borrowed books, fines
Hostel	Room booking, gate access
Campus Wi-Fi	Network access, device registration

You do **not** need to log in again — each service independently verifies your ABU-SSO token.

> **First-click delay:** On the free-tier deployment, the first click after 15 minutes of inactivity may take 30–60 seconds while the service wakes up. Subsequent clicks are instant.

A.9 Logging Out

- Click the red **Log out** button on the dashboard.
- Your token is deleted from the browser.
- You are redirected to the login page.
- If you try to open `/dashboard` without logging in, you will be sent back to `/login`.

A.10 Troubleshooting

Problem	Likely cause	Solution
"Invalid matric/staff number or password"	Wrong credentials	Re-type carefully; matric is not case-sensitive
"Not authenticated"	Token expired (15 min)	Log in again
Dashboard shows "—" or blank	Token missing	Log out and log back in
Service card opens but page times out	Free-tier service sleeping	Wait 30–60 s and refresh
"Invalid or expired token" on a service	Token expired between clicks	Return to dashboard, log out, log back in, click immediately
Page looks unstyled	Cache issue	Press **Ctrl + F5**

Part B — Installation Guide (For Developers)

B.1 Prerequisites

Tool	Version	Purpose
Python	3.12.x	Backend runtime
Git	2.40+	Version control

Tool	Version	Purpose
VS Code	Latest	Code editor
Anaconda or Miniconda	Latest	Environment management

B.2 Clone the Repository

git clone https://github.com/idrismakr-png/abu-sso.git cd abu-sso B.3 Create and Activate the Environment

conda create -n abu-sso python=3.12 -y conda activate abu-sso B.4 Install Dependencies

python -m pip install --upgrade pip pip install -r requirements.txt B.5 Configure Environment Variables (Optional) Copy .env.example to .env and adjust:

copy .env.example .env # Windows cp .env.example .env # macOS/Linux Edit .env and set a strong JWT_SECRET (at least 32 characters).

B.6 Run the Application

uvicorn app.main:app --reload Then open:

Dashboard: http://127.0.0.1:8000/login

API docs: http://127.0.0.1:8000/docs

B.7 Run the Six Mock SSO Services (optional) In six separate terminals:

cd mock-services/portal && uvicorn main:app --reload --port 4001 cd mock-services/lms && uvicorn main:app --reload --port 4002 cd mock-services/library && uvicorn main:app --reload --port 4003 cd mock-services/hostel && uvicorn main:app --reload --port 4004 cd mock-services/wifi && uvicorn main:app --reload --port 4005 cd mock-services/results && uvicorn main:app --reload --port 4006 The dashboard auto-detects localhost vs. production and wires the cards accordingly.

B.8 Run the Tests pytest -q # 79 fast tests pytest tests/e2e -m e2e # 5 E2E tests Expected: 84 passed.

B.9 Optional — Coverage Report

pip install pytest-cov pytest --cov=app --cov-report=html Open htmlcov/index.html in your browser.

B.10 Optional — Static Security Scan

pip install bandit bandit -r app/ Expected: 0 findings.

B.11 Optional — Deploy to Render The repo includes render.yaml, which declares all seven web services (IdP + 6 mock services). To deploy:

Push your repo to GitHub.

Go to https://render.com → New → Blueprint → connect your repo.

Render detects render.yaml and creates all seven services.

Set the DATABASE_URL environment variable on the abu-sso service to your PostgreSQL connection string (e.g., from Neon).

Wait ~5 minutes for all services to build.

Your IdP is live at https://abu-sso.onrender.com.

Part C — Administrator Guide C.1 Getting Admin Access The first-ever user on a fresh deployment becomes admin automatically. This is called the "bootstrap" user.

To create the bootstrap admin (fresh deployment only):

Visit https://abu-sso.onrender.com/docs

Find POST /auth/register → click "Try it out"

Submit a JSON body:

json { "email": "admin@abu.edu.ng", "password": "admin123", "full_name": "System Administrator", "role": "student", "matric_no": "ADMIN/001" } The response shows "role": "admin" — the bootstrap worked.

After this, POST /auth/register returns HTTP 403 — no more public registrations.

C.2 Logging In as Admin Go to /login

Enter matric/staff number: ADMIN/001

Enter password: admin123

You'll see the dashboard with a blue Admin Panel button.

Click Admin Panel to reach /admin.

C.3 The Admin Panel The Admin Panel has four sections:

1. System Statistics

Total users, active users, wallets, total balance, total transactions.

2. Create User Account

Full Name, Email, Matric / Staff Number, Role (student / staff / admin), Initial Password.

The new user can log in immediately with their matric number and password.

All fields are required.

3. Top Up Any Wallet

Enter a user's matric number and an amount.

Instantly credits their wallet with a "credit" transaction.

4. All Users

Table of every account with columns: Matric, Name, Email, Role, Active, Actions.

Change role: select from the dropdown — saves immediately.

Activate / Deactivate: toggle a user's account. Deactivated users cannot log in.

C.4 Admin Operations — Step by Step Create a new student:

Scroll to "Create User Account"

Fill in: Jane Doe, jane@abu.edu.ng, P25EGCP1234, student, welcome123

Click Create Account

Green confirmation appears; Jane can now log in with P25EGCP1234 / welcome123

Promote a user to staff:

In the "All Users" table, find the user

Change the Role dropdown from student to staff

Confirm the notification

Deactivate a user:

In the "All Users" table, find the user

Click the red Deactivate button

Their is_active becomes false; login attempts are rejected

Top up a wallet:

Scroll to "Top Up Any Wallet"

Enter the user's matric number and amount (e.g., 1000)

Click Top Up

C.5 Rotating the JWT Secret Change JWT_SECRET in .env (local) or in Render's Environment tab (production).

Restart the application (Render does this automatically).

All existing tokens become invalid — every user must log in again.

C.6 Backups SQLite (dev): copy abu_sso.db to a safe location.

PostgreSQL (production): use Neon's built-in PITR (point-in-time recovery).

C.7 Monitoring Ping /health every 60 seconds (UptimeRobot or similar).

If it fails, check the Render logs.

Part D — Support For issues related to the ABU-SSO prototype, contact the project author:

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